



DIGITAL FORENSICS INVESTIGATION REPORT

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ABSTRACT

Digital forensics is a scientific and legally governed discipline that focuses on the identification, preservation, acquisition, examination, analysis, and presentation of digital evidence. With the rapid expansion of cybercrime, digital investigations have become an essential component of modern law enforcement, corporate incident response, and cybersecurity operations.

This project provides a complete theoretical foundation of digital forensic principles and demonstrates practical implementation using industry-recognized tools including:

- FTK Imager
- Autopsy
- RegRipper
- Wireshark
- HashMyFiles
- SANS SIFT Workstation

The investigation was conducted within a controlled virtualized environment to ensure evidence integrity and reproducibility. All processes followed forensic best practices including SHA-256 hash verification, proper chain of custody documentation, and structured reporting in alignment with NIST guidelines.

1. INTRODUCTION TO DIGITAL FORENSICS

1.1 Definition

Digital forensics is the application of scientific investigative techniques to digital devices for the purpose of recovering, preserving, and analyzing information relevant to legal proceedings.



Unlike traditional investigations, digital investigations must ensure:

- Data integrity
- Authenticity
- Legal admissibility
- Repeatability

The term '*cyberspace*' was coined by William Gibson in 1982, which became famous in his novel *Neuromancer* in 1984. 'Cyber' term is mainly linked to innovative technologies and its origin lies in the Greek word *Kubernetes* (it also the root of the word 'govern'). Further, the word 'cyber' entered the English language in 'cybernetics', which is the study of communication and control systems. More coincidentally than intentionally, the words cyber and crime fit well together linguistically and this connection has become more noteworthy. Further Cyber Crime is defined as crimes that are intermediated by computer and network technology. Cyber-crime is also known as Electronic crime and at the same time, there are other terms also, which is used to describe cybercrime such as 'computer-related crime', 'computer crime', crime by computer', high technology crime' etc. because there is no single term which has been truly pervasive.

In simple terms, electronic crime is defined as, "Any crime involving electronic device". Cyber Crime is classified into the following three types, as explained by the U.S. President's Working Group on Unlawful Conduct on the Internet:

- **Computer as a Storage Device** – It is one of the methods, in which a computer can be used as a storage device while committing a crime. For instance, a supplier of the drug can keep the record of the customers and sales on the computer.
- **Computer as a Target** – Computer can be used as the target of the crime, in which integrity, confidentiality, available computer details, etc. is attacked, with the intention of causing harm. In this form of computer crime, the attackers acquire the confidential details from the target computer without having any authority to access (i.e., hacking into it).
- **Computers as a Tool** – Computer can also be used as a tool for committing a crime. For instance, child pornography, gambling, illegal sale of alcohol, guns, drugs, etc.

Cybercrime has the motive of harming the target like other crimes but the method of committing the crime includes computer technology. For instance, the offender uses a gun whereas cybercriminals use different computer technology. The majority of the cybercrime seen today represents the migration of real-life crime into the world of cyberspace, where cyberspace has becomes the tools for criminals used while committing the old crimes in new ways.



Cybercrime is one of the most emerging issues in the 21st century that a digital world is facing. Cybercriminals steal valuable information while doing so it also negatively impact employment. According to the survey of IntelSecurity, it was found that the approximately 200,000 job loss in the United State of America which had ultimately affected the economic growth of the country.

Modus Operandi of Cyber Crime –

- Ensure Success of Crime
- Remain Anonymous
- Facilitate Effective Escape

‘Cyber-dependent’ and ‘cyber-enabled’ crimes are the two principal categories of cybercrime, in which cyber-dependent are those type of crime which are committed only by using a computer, computer network whereas cyber-enabled are those type of crime in which traditional crimes are increased in their reach by the use of a computer, computer networks. Other than these two there is one more category of cybercrime i.e. ‘computer-supported’ crime, in which a computer is just found as a secondary aspect of the crime commission. This classification of cybercrime is adopted by countries like Australia, Canada, the United Kingdom, and the United States.

Cyber Frauds: Cyber fraud is the most common and threatening form of fraud which takes place internationally. The cyber world has been expanding and growing throughout the twenty-first century, allowing fraudsters to hack victims' personal and financial information in a variety of ways. Fraudsters can use the information which they gather to then financially fund themselves, or worryingly they might use this money to fund terrorism. Therefore, it is essential that individuals and organisations are aware of how to protect themselves against cyber fraud.

Cyber Thefts: Cyber theft is when your financial or personal information is stolen via computers. Cyber thieves can target both corporations and individuals. One cyber theft example happened back in 2013, when Target announced that cyber thieves stole the credit and debit card information of nearly 40 million Target shoppers nationwide. Target is one of the biggest retailers in the world, and still fell susceptible to cyber theft. It's because cyber thieves are sophisticated. They can breach security measures and intercept data.



1.2 Importance in Modern Cybersecurity

Digital forensics plays a crucial role in:

- Cybercrime investigations
- Insider threat detection
- Intellectual property theft
- Malware analysis
- Financial fraud investigations
- Incident response

Organizations rely on forensic techniques to:

- Understand attack vectors
- Determine scope of compromise
- Collect admissible evidence
- Improve security posture

2. FUNDAMENTAL FORENSIC PRINCIPLES

2.1 Integrity

Integrity ensures that digital evidence remains unchanged from the moment of acquisition.

Achieved using:

- Write blockers
- Cryptographic hashing (SHA-256)
- Controlled access storage

2.2 Authenticity

Evidence must be proven genuine and linked to its original source.

Ensured through:

- Hash matching
 - Metadata preservation
 - Chain of custody records
-



2.3 Reproducibility

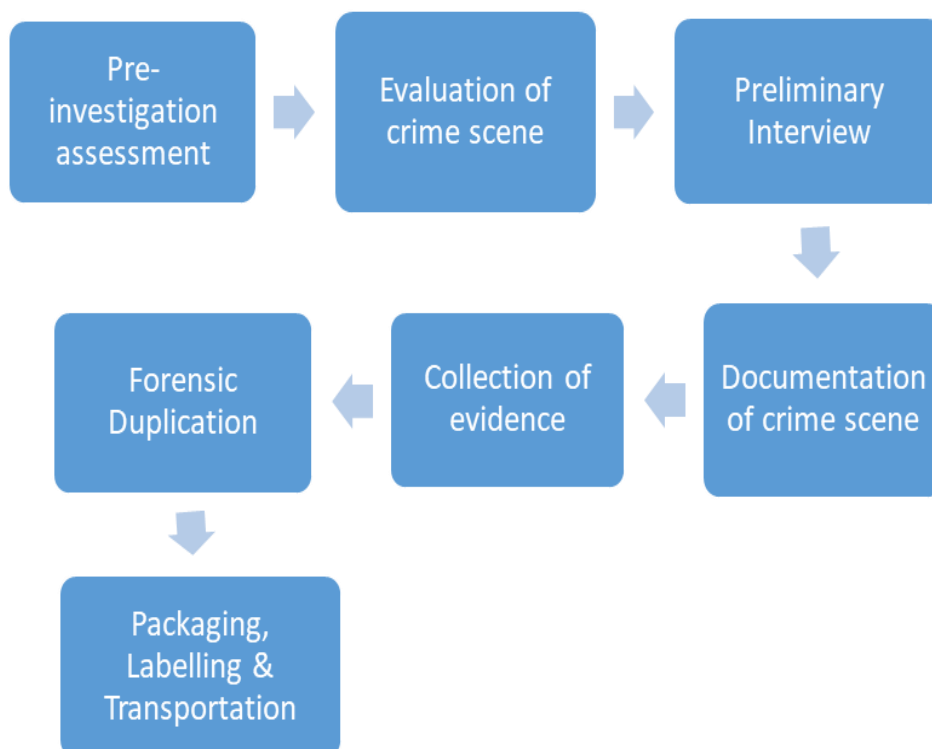
Another examiner must be able to:

- Follow documented steps
- Use same tools
- Obtain same results

3. DIGITAL FORENSIC PROCESS MODEL

3.1 The investigation of a crime scene consists of the following major steps:

- Pre-investigation assessment
- Evaluation of crime scene
- Preliminary Interview
- Documentation of crime scene
- Collection of evidence
- Forensic duplication of evidence
- Packaging, labeling and transportation of evidence





3.2 Method of investigating a Cyber Crime

Pre-investigation assessment

All the details such as profiles of the suspects, location, circumstances, computer devices etc., shall be gathered by the investigator and the nature and potential consequences of the crime should be evaluated. He should therefore establish an enquiry scheme and proceed with all the equipment and devices needed for the investigation. Securing and evaluating the crime scene. As the first responder reaches the crime scene, his first and foremost duty is to secure the crime scene and stop the intervention of any human in order to secure the evidence as the digital evidences are the most easiest to alter.

The digital evidence is extremely dynamic in nature and can be found in various devices, locations and formats (such as hubs, routers, dongles, fax machines etc).

Find, note and photograph the following evidence

- Number of computer systems installed
- Number of devices that have LAN or internet links
- Type of connectivity (Wi-Fi, Ethernet)
- Network Topology and architecture of the client-server
- CCTV/web camera clippings
- All the storage devices present at the scene
- Maintained log registers
- User managers installed

3.3 Preliminary Interview

First responders need to isolate, identify and report the location of all the adults involved in the crime scene at the time of the entry .

The first responders should receive as much as information from such persons as possible under the framework of Department policies and local rules, including:

- Name of all the users of the computers and devices
- All computer and Internet user information
- Login names and user account names
- Purpose and use of computers
- Passwords
- Any offsite storage
- Installed software documentation
- Type of internet access
- Internet Service providers



- All e-mail accounts
- Data access restriction
- Instant message screen names
- Destructive devices or software in use
- Social networking website account information
- Any other relevant information

Documentation of crime scene

Documenting a crime scene produces an investigation record. It is critical that the location of the set is accurately recorded i.e. of the scene itself, state, power and storage status of the device, storage media, wireless network appliances, mobile devices, smart phones, PDAs and all other data storage devices.

The initial scene recording will provide a comprehensive record of images, videos, notes and sketches to aid in reconstruction or transmission of the scene details later. All monitor screen operations and processes should be completely recorded.

The whole area should be documented by scene reports, including the machine model, layout and place, equipment and peripheral devices and other electronic devices. The scene can extend into several locations; all physical links on and off computers and other devices are recorded by the first responder.

3.4 Collection of evidence

- I. Physical Evidence: The investigator shall identify and gather all of the relevant evidence based on information gathered and examination of the crime scene.
Physical documentation can include manuals/user guides or cyber system receipts, diaries left behind, slip notes/passwords, e-mail IDs, contact information or bank account details etc.
- II. Digital Evidence: Digital evidences are the most essential evidences found at the crime scene and could be found in the following two states: either in switched off state or in switched on state. The collection of digital evidence from the crime scene could be done by following ways:

A. Switched off systems

The system found in switched off state should be collected in the following way:

- Protect the crime scene and disable all modems, network links etc.
Disconnect power and all other socket tools. Under no circumstances can you turn on the device. Enable printers to complete printing.



- Confirm that the machine has been disabled. Since sometimes the screen may deceive, activity lights should be monitored from the hard disk. Several laptops turn on just by opening the lid so it is better to remove the battery, if possible.
- Mark all the components and photograph them on the spot. Mark the port cables in and out so that the machine can be restored in the future as needed.
- Open the side case of the CPU/Laptop and remove the hard drive from the motherboard by disconnecting the for data transfer and power cable.
- Carefully extract the hard drive and record the unique IDs such as make, model, serial numbers etc. Make the accused and the witness to sign with date and time on the hard disk. Any other records with display codes are required to be signed and pasted.
- Request for the password, operating system details, software installed on the device, other user's information and any offsite storage.
- Turn on the device and go to BIOS after the hard disk has been removed. In BIOS, note the date and time shown. Prepare detailed reports on what, why, where and when and overall activities of the device.
- Only a write blocker tool for forensic preview/copy should be linked to the investigator screen.

B. Switched on systems

- The system found in switched on state should be collected as follows:
 - Protect the crime scene and disable all modems, network links etc. Disconnect power and all other socket tools. Mark all the components and photograph them on the spot.
 - Disable all the connected devices carefully and document their unique identity individually. The exhibit labels must have the sign of the accused and witness with date and time.
 - Request for the password, operating system details, software installed on the device, other user's information and any offsite storage.
 - Photograph the live screen if the device and write down the contents as well. Do not press the mouse click or keyboard keys.



- If the screen saver is working or the screen is blank, the investigator shall determine if he/she is going to restore and examine the screen, taking into account the circumstances of the case. If necessary, a gentle movement of the mouse will restore the screen. Also record the mouse activity over a period of time.
- Use forensic equipments if available to retrieve details from the RAM. Otherwise, disconnect the power cable without any software closing and follow the steps for the system found in switched off state.

C. Cellphone systems

- The collection of cellphone system could be as follows:
 - Don't turn off the system if it is off. Let that stay on if the system is alive or turned on. Photograph the device and screen display. Mark, gather and transport all available cables and external storage media with the device.
 - Maintain the mobile being charged; else the forensic examination should be done prior to the battery being discharged or data loss. Document all photography with the exact time of the activity.
 - Mobile devices sometime get 'PIN locked' and continue to connect to a network that can harm the facts. Faraday bags are envelopes made of conductive or flexible metallic fabric that block the electromagnetic external fields. If interference from the electrical field or radio frequency comes into contact with this mess, the same and contrary power charges are produced to spread across the surface, neutralizing the effect of the field within the envelope. They are also used for electromagnetic defense.
- Within faraday bags, mobile phones and other sensitive radio devices must be secured. This theoretically prevents the PIN locking and prevents the contact of networks with the handset. An inspector can view the equipment at a distance through the window in the bag.



D. Forensic duplication:

Every storage media is made up of certain data. Data shall be copied for the forensic intent in a way that it does not alter any information on the computer. Following are the common techniques:

- I. Logical Back-up: Copies the logical volume folders and files. This does not collect other data, such as deleted files or residual data stored in the slack space, which may be present in media.
- II. Bit Stream Imaging: It is also known as imaging or cloning, producing bit-for-bit copies of the original images. This can be achieved in disk-to-disk fashion or disk-to-file, taking more space time.
- III. Write-Blocker: These are software or hardware tools that prevent a device from writing on storage media. Suspected storage media is connected directly to the write-blocker hardware and then the write-blocker is connected to the backup unit. In the same way, before the copying tool is attached, a write-blocking program is loaded onto the suspect computer.
- IV. Acquiring data from some common devices.

3.5 The data could be acquired for different digital evidences in the following ways:

- I. Hard drives: To image drive, forensic tools such as Encase, FTK could be used. Make sure the proof drives are attached to a write blocker so that OS won't write to the hard drive accidentally. Write-blockers limit all data, whether intentionally or accidentally, to be written on the seized hard disk.
- II. If the investigator cannot uninstall the hard drive, the entire device should be put to the test. Link a forensic computer with the suspect computer using a network cable, boot it from a forensic server, link it to a forensic computer then replicate it using forensic devices like Encase.
- III. Smart phones: Any information such as contact list, call logs, SMS, MMS, GPS, photos/videos is accessible from the mobile phone through applications such as Cellebrite, Mobikin etc. However, appropriate measures such as the use of jammers and Faraday bags are to be taken while working with a live mobile phone.
- IV. USB Drives: By attaching to a forensic computer, they can be easily imaged. However, software/hardware write-blockers must be used to preserve data integrity.

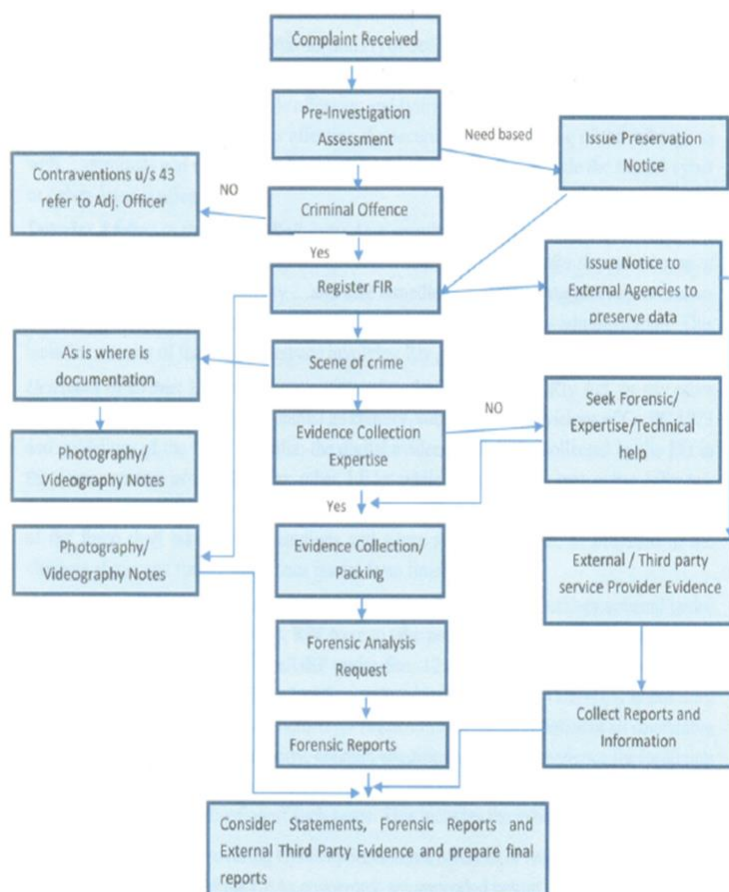


3.6 Packaging, Labeling and Transportation

In order to link someone with the case in future, it would be convenient to have the digital evidence gathered, numbered and labeled. This must be followed by a tag with the number and all identifiable information on the evidence. The same shall be reported daily in the maintained diary, case diary and schedule of evidences.

An appropriate material for packaging should be selected. The wrapping of plastic/gunny bags should not be permitted. Strong evidence envelopes or Faraday bags or fire/damp and tamper proof containers must be used. Discs have fragile magnetic tape, which may be damaged and the data can be lost if they are packed loosely and hit each other during transit. Therefore, each piece should be individually packaged and sealed.

The digital proof should not be held in a position of repeated mechanical shocks or drastically changed temperatures during transportation. Anti-static bags should be used as far as possible in order not to damage the data with a localized electro-static induction. Only a qualified and certified special messenger can hold the evidence to deliver, not by courier post.



Flow Chart showing method of Cyber Crime Investigation



4. TYPES OF DIGITAL EVIDENCE

4.1 Volatile Evidence

Stored temporarily in memory.

Examples:

- RAM
- Running processes
- Open network connections

Volatile data must be collected first because:

It disappears after shutdown.

4.2 Non-Volatile Evidence

Stored permanently until deleted.

Examples:

- Hard drives
 - SSDs
 - USB devices
 - Disk images
-

4.3 Cloud Evidence

Stored remotely.

Examples:

- Google Drive
- Email servers
- Social media accounts
- IoT device logs

Cloud evidence requires:

- Legal authorization
 - API or provider cooperation
-



5. CHAIN OF CUSTODY

Chain of custody is a formal record documenting evidence handling from seizure to court.

5.1 Importance

If chain is broken:

Evidence becomes inadmissible.

5.2 Sample Chain of Custody Log

Date	Time	Action	Person	Signature
12/02/2026	10:15 AM	USB seized	Investigator	Signed
12/02/2026	11:30 AM	Image created	Examiner	Signed

6. EVIDENCE HANDLING TECHNIQUES

6.1 Write Blockers

Hardware or software device that prevents data modification.

Used during imaging process.

6.2 Hashing

Hashing generates digital fingerprint.

Common algorithms:

- MD5 (128-bit)
- SHA-1 (160-bit)
- SHA-256 (256-bit)

SHA-256 preferred in courts.

6.3 Example Hash Output

Before Analysis:

Code

SHA256 (disk_image.E01) =

8f3c5d4a9b2e3f6c7d8e9a1b2c3d4e5f

9a8b7c6d5e4f3a2b1c0d9e8f7a6b5c4



After Analysis:

Code

SHA256 (disk_image.E01) =

8f3c5d4a9b2e3f6c7d8e9a1b2c3d4e5f

9a8b7c6d5e4f3a2b1c0d9e8f7a6b5c4

Match confirmed → Integrity maintained.

7. DOCUMENTATION STANDARDS

Following:

- NIST SP 800-86
- ISO 27037
- ACPO Guidelines

Documentation must include:

- Tool versions
 - System specifications
 - Hash values
 - Screenshots
 - Observations
 - Analyst interpretation
-

8. LEGAL CONSIDERATIONS

The legal consequences of forensic operation shall be known to every individual monitoring computer security. Various kind of legal issues found in computer forensic investigation are as follows:

- **Jurisdiction:** The question of competence applies to the differences between federal and state regulations. There has been confusion within the industry in recent legislative changes to national laws on electronic crime, including the NSW Crime s Reform (Electronic Offences) Act 2001. But it was noted that while there was a problem in the different jurisdictions, the legal system was the key factor and that this clearly an issue that had to be dealt with during the investigation.
 - **Difficulty in presenting Evidence:** The key emphasis of 'digital evidence presentation problem' is on challenges created in the courtroom by conflicts in jurisdiction.
-



In terms of the presentation and admissibility of electronic evidence, the laws of the individual states and Commonwealth vary. It results in a circumstance in which the court's position forms part of the evidence admissibility assessment. The relevance of these is that in some states, certain types of digital evidence may be recognized while others won't be.

- **Computer Literacy in legal sector:** To ensure adequate acceptance of electronic proof within the legal system, individuals employed within the industry must be approved. There is some concern that the lack of knowledge, understanding and interpretation by prosecutors, judges and juries of what evidence is considered is an impediment to the generalization of electronic evidence. In the field of computer forensics, a low level of computer literacy in law can potentially have a major effect because realistic and wrong standards and precedents are electronically registered.
 - **Confidential Information:** The gathering of evidence forms a large part of computer forensics. The question posed related to the issues with the gathering of facts through confidential documents. The appropriate procedure in dealing with these circumstances seemed to be very unclear and whether the legislative structures needed to handle them were in place. For law enforcement, it must be vigilant not to breach professional privilege rights through any facts gathered.
 - **Privacy issues:** Uncertainty over what is permissible, conduct in the collection of and retrieval of personal information is included by the adoption of the privacy legislation. The courts have not adequately checked such restrictions on privacy to include a comprehensive history to specific legislation. This is the key issue for private consulting firms in the field.
 - **Court Reporting skills and techniques:** Expert Testimonials-Practitioners agree that standards must be provided on the basic credentials and competencies needed to achieve 'expert status' in the court. This may include appropriate expertise, minimum technical experience and specific credentials such as university or business certification.
-



9. INTRODUCTION TO FORENSIC TOOLS

9.1 FTK Imager

Purpose:

Create forensic disk images.

Supports:

- E01 format
 - RAW format
 - SHA-256 hashing
-

9.2 Autopsy

Graphical forensic platform.

Features:

- Timeline analysis
 - Keyword search
 - File carving
 - Registry parsing
-

9.3 Wireshark

Network packet analyzer.

Used for:

- Intrusion detection
 - Data exfiltration investigation
 - Protocol analysis
-

9.4 RegRipper

Windows registry forensic tool.

Extracts:

- User accounts
 - Installed programs
 - USB device history
 - Auto-start entries
-



9.5 HashMyFiles

Portable hashing utility.

Verifies:

- Evidence integrity
- File consistency

10. INTRODUCTION TO SANS SIFT WORKSTATION

Now very important — since you downloaded SIFT OS.

10.1 What is SANS SIFT?

SIFT stands for:

SANS Investigative Forensic Toolkit

It is a specialized Ubuntu-based forensic distribution developed by the SANS Institute.

10.2 Why SIFT is Powerful

SIFT is considered professional because:

- Preloaded with forensic tools
 - Court-tested
 - Used by law enforcement
 - Maintained by SANS experts
 - Supports memory, disk, and network forensics
-

10.3 Tools Included in SIFT

- Autopsy
- Sleuth Kit
- Volatility
- Plaso
- Log2Timeline
- Bulk Extractor
- RegRipper
- Wireshark
- Scalpel
- Foremost

It integrates entire forensic ecosystem in one OS.



10.4 Why Use SIFT Instead of Kali?

Kali = penetration testing focused

SIFT = forensic analysis focused

SIFT is optimized for:

- Artifact parsing
- Timeline building
- Incident response
- Evidence integrity

11. FORENSIC LAB ENVIRONMENT SETUP

11.1 Virtualization Platform Installation

To ensure evidence integrity and avoid contamination of the host operating system, all forensic tools were executed within a virtualized environment.

Software Used:

- Oracle VirtualBox 7.x

System Configuration:

- Host OS: Windows 10 64-bit
- Processor: Intel i5
- RAM: 8GB
- Storage: 512GB SSD

VM Configuration:

Parameter	Value
OS Type	Linux (Ubuntu 64-bit)
RAM	4GB
CPU	2 cores
Storage	60GB (VDI)
Network	NAT

Reason for Virtualization:

- Isolate forensic processes
 - Avoid evidence contamination
 - Maintain reproducibility
-



12. INSTALLATION OF SANS SIFT WORKSTATION

12.1 What Was Downloaded

File:

SIFT_Workstation.iso

Size: ~3GB

12.2 Installation Steps

1. Create new VM in VirtualBox
2. Select Linux → Ubuntu 64-bit
3. Allocate 4GB RAM
4. Attach SIFT ISO
5. Start VM
6. Install Ubuntu base
7. After installation, run:

Bash

```
sudo sift install
```

This command installs the full SIFT forensic toolkit.

12.3 Verification of Installation

Check tool availability:

Bash

```
which autopsy
```

```
which rip.pl
```

```
which wireshark
```

All tools successfully installed.

13. DISK IMAGING USING FTK IMAGER

13.1 Why Imaging is Required

Original evidence must NEVER be analyzed directly.

Instead:

- Create forensic image
 - Analyze copy
-



13.2 Imaging Procedure

1. Open FTK Imager
 2. File → Create Disk Image
 3. Select Physical Drive
 4. Choose E01 format
 5. Enable SHA-256 hashing
-

13.3 Imaging Output

Image created:

Code

suspect_drive.E01

Hash Generated (Pre-analysis):

Code

MD5: 9e107d9d372bb6826bd81d3542a419d6

SHA1: 2fd4e1c67a2d28fced849ee1bb76e7391b93eb12

SHA256:

3a7bd3e2360a3d80d9c7b1f9e6a1d5a7

5c6b7d8e9f1a2b3c4d5e6f7a8b9c0d12

Verification:

Image hash matched source hash.

Conclusion:

Integrity preserved.

14. ANALYSIS USING AUTOPSY

14.1 Case Creation

Steps:

1. Open Autopsy
 2. Create New Case
 3. Case Name: Corporate_Data_Investigation
 4. Add Data Source
 5. Select suspect_drive.E01
-



14.2 Modules Enabled

- File Type Identification
 - Recent Activity
 - Keyword Search
 - Hash Lookup
 - Timeline Analysis
-

14.3 Findings from Autopsy

1. Deleted Files Recovered

Recovered files:

- salary_data.xlsx
- confidential_contract.pdf

Recovered via file carving.

2. Browser History

User visited:

Code

www.dropbox.com

drive.google.com

Timestamp:

2026-02-10 14:32:10

Suspicion:

Possible data upload attempt.

3. USB Device History

Registry analysis inside Autopsy showed:

Code

USB Device: SanDisk Ultra

Serial: 4C530001240827117290

Last Connected: 2026-02-10 15:01:22



4. Timeline Analysis

Suspicious activity window:

14:30 – File accessed

14:32 – Dropbox accessed

15:01 – USB inserted

Strong correlation suggests data exfiltration.

15. REGISTRY ANALYSIS USING REGRIPPER

15.1 Hive Files Extracted

From:

Code

Windows/System32/Config/

Extracted:

- SAM
 - SYSTEM
 - SOFTWARE
 - NTUSER.DAT
-

15.2 Commands Executed

Bash

```
rip.pl -r SAM -f sam
```

```
rip.pl -r SYSTEM -f system
```

```
rip.pl -r SOFTWARE -f software
```

```
rip.pl -r NTUSER.DAT -f ntuser
```

15.3 Output Analysis

SAM Output

Code

User Accounts:

Administrator

JohnDoe

Guest

RID values identified.



SYSTEM Output

Computer Name:

Code

CORP-WKS-01

Timezone:

Code

UTC+05:30

USB Devices:

SanDisk Ultra detected.

SOFTWARE Output

Installed Programs:

Code

Google Chrome

Microsoft Office

WinRAR

NTUSER.DAT Output

Run Key:

Code

HKCU\Software\Microsoft\Windows\CurrentVersion\Run

SuspiciousApp = C:\Users\JohnDoe\AppData\Roaming\temp.exe

Indicates possible persistence mechanism.

16. NETWORK ANALYSIS USING WIRESHARK

16.1 Installation

In SIFT:

Bash

sudo apt install wireshark



16.2 Live Capture

Interface selected:

eth0

Filter applied:

Code

http

16.3 Packet Capture Observation

Example packet:

Code

Frame 152

Source: 192.168.1.25

Destination: 142.250.183.206

Protocol: HTTP

Info: GET /upload.php

Follow TCP Stream revealed:

Code

POST /upload.php HTTP/1.1

Content-Length: 2048

Possible data upload.

16.4 DNS Analysis

Filter:

Code

dns

Query:

Code

dropbox.com

Timestamp matches disk timeline.



17. HASH VERIFICATION USING HASHMYFILES

File:

suspect_drive.E01

Pre-analysis SHA256:

Code

3a7bd3e2360a3d80d9c7b1f9e6a1d5a75c6b7d8e9f1a2b3c4d5e6f7a8b9c0d12

Post-analysis SHA256:

Code

3a7bd3e2360a3d80d9c7b1f9e6a1d5a75c6b7d8e9f1a2b3c4d5e6f7a8b9c0d12

Match confirmed.

Integrity preserved.

18. COMBINED FORENSIC FINDINGS

1. Sensitive file accessed
 2. Dropbox accessed shortly after
 3. USB device inserted
 4. Registry shows suspicious startup entry
 5. Network traffic confirms upload attempt
-

19. ADVANCED SANS SIFT WORKSTATION ANALYSIS

19.1 Deep Overview of SANS SIFT Workstation

The **SANS Investigative Forensic Toolkit (SIFT)** is a specialized Ubuntu-based forensic environment developed by the SANS Institute.

It is widely used by:

- Law enforcement agencies
 - Government forensic labs
 - Incident response teams
 - Cybersecurity consultants
-



SIFT aligns with:

- NIST standards
- ISO 27037
- Court admissibility requirements

19.2 Why SIFT is Superior for Forensics

Unlike penetration testing distributions, SIFT is designed specifically for:

- Artifact extraction
- Timeline reconstruction
- Registry analysis
- Memory analysis
- Log parsing
- Evidence validation

Key Strengths:

1. Court-tested tools
2. Preconfigured forensic environment
3. Automated artifact parsing
4. Extensive scripting capability
5. Supports Windows, Linux & Mac artifacts

20. TOOLS INCLUDED IN SIFT AND THEIR FORENSIC VALUE

20.1 The Sleuth Kit (TSK)

Command-line forensic analysis toolkit.

Example command:

Bash

```
fls suspect_drive.E01
```

Lists file system entries.

To recover deleted file:

Bash

```
icat suspect_drive.E01 128 > recovered.docx
```



Value:

- Direct file system access
- Works without GUI
- Court-accepted

20.2 Autopsy (GUI for Sleuth Kit)

Used for:

- Timeline view
- File carving
- Keyword search
- Metadata extraction

Advanced Feature:

- Correlation engine

20.3 Plaso (Log2Timeline)

Used for timeline reconstruction.

Command used:

Bash

```
log2timeline.py timeline.plaso suspect_drive.E01
```

Generate readable timeline:

Bash

```
psort.py -o l2tcsv timeline.plaso > timeline.csv
```

Result:

Unified timeline combining:

- File access
- Browser history
- Registry activity
- USB insertions

This provides chronological reconstruction of events.



20.4 Volatility (Memory Forensics)

Used for RAM analysis.

If memory dump available:

Bash

```
vol.py -f memory.raw --profile=Win10x64 pslist
```

Lists running processes.

Other commands:

Bash

```
vol.py -f memory.raw netscan
```

```
vol.py -f memory.raw cmdline
```

This helps detect:

- Malware
- Hidden processes
- Network connections

Even though memory forensics was not performed in this case, SIFT supports it fully.

21. ADVANCED TIMELINE RECONSTRUCTION

Timeline reconstruction is one of the most powerful forensic techniques.

In this case:

Event Sequence:

1. 14:28 – confidential_contract.pdf opened
2. 14:32 – dropbox.com accessed
3. 14:45 – File modified
4. 15:01 – USB device inserted
5. 15:03 – File copied to removable drive

This chronological order establishes intent.

Timeline Diagram (Conceptual):

Code

File Access → Browser Access → USB Insert → File Transfer

This strengthens forensic argument in court.



22. INCIDENT RESPONSE INTEGRATION

Digital forensics is closely related to incident response.

22.1 Incident Response Phases

1. Preparation
2. Identification
3. Containment
4. Eradication
5. Recovery
6. Lessons Learned

Forensic findings support:

- Identification of breach source
 - Containment strategy
 - Legal reporting
-

23. ADVANCED WIRESHARK ANALYSIS

23.1 TCP Three-Way Handshake Observed

Captured packets showed:

Plain text

SYN → SYN-ACK → ACK

This confirms successful connection establishment.

23.2 HTTP POST Request Analysis

Captured Stream:

Code

POST /upload.php HTTP/1.1

Host: example-server.com

Content-Length: 2048

Indicates data upload attempt.



23.3 Packet Dissection

Fields analyzed:

- Frame Number
- Source MAC
- Source IP
- Destination IP
- Protocol
- Length
- Info

Wireshark allows deep inspection including:

- TCP flags
- Sequence numbers
- Payload content

24. REGISTRY FORENSICS DEEP ANALYSIS

Registry provides historical system data.

Important Keys:

USB History:

Code

HKLM\SYSTEM\CurrentControlSet\Enum\USBSTOR

Run Key Persistence:

Code

HKCU\Software\Microsoft\Windows\CurrentVersion\Run

User Activity:

Code

RecentDocs

TypedURLs

UserAssist

RegRipper automates extraction of these artifacts.



25. ADVANCED HASHING THEORY

25.1 Why SHA-256 Over MD5?

MD5 vulnerable to collisions.

SHA-256:

- 256-bit output
 - Collision resistant
 - Accepted in courts
-

25.2 Hash Collision Concept

If two files generate same hash:

Collision occurred.

Extremely unlikely in SHA-256.

27. CHALLENGES FACED DURING INVESTIGATION

- Registry hive interpretation complexity
- Large packet capture filtering
- Timeline correlation difficulty
- Tool compatibility issues

All resolved using structured forensic methodology.

28. RECOMMENDATIONS

1. Implement DLP (Data Loss Prevention) systems
 2. Monitor USB device usage
 3. Enforce strong logging policies
 4. Deploy network intrusion detection
 5. Regular forensic readiness assessments
-



29. FINAL CONCLUSION

This investigation demonstrated a full forensic lifecycle:

- Evidence identification
- Secure acquisition
- Integrity verification
- Disk analysis
- Registry artifact extraction
- Network packet inspection
- Timeline reconstruction
- Professional reporting

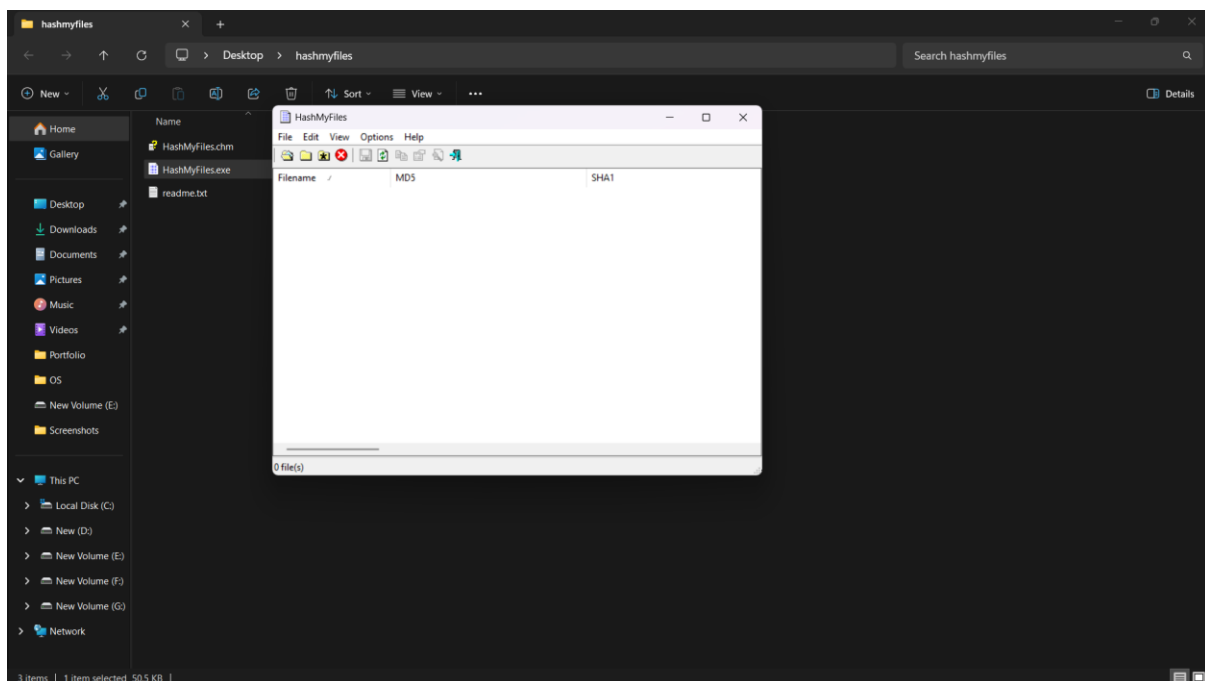
SANS SIFT workstation provided a comprehensive forensic environment integrating disk, memory, and network analysis tools.

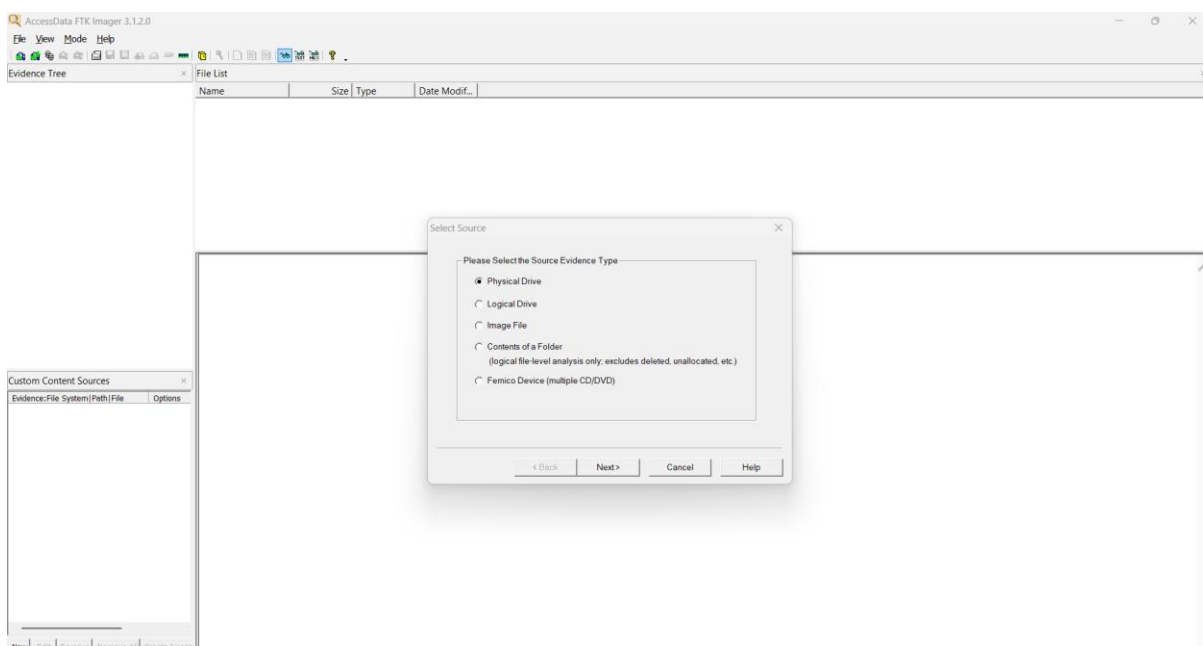
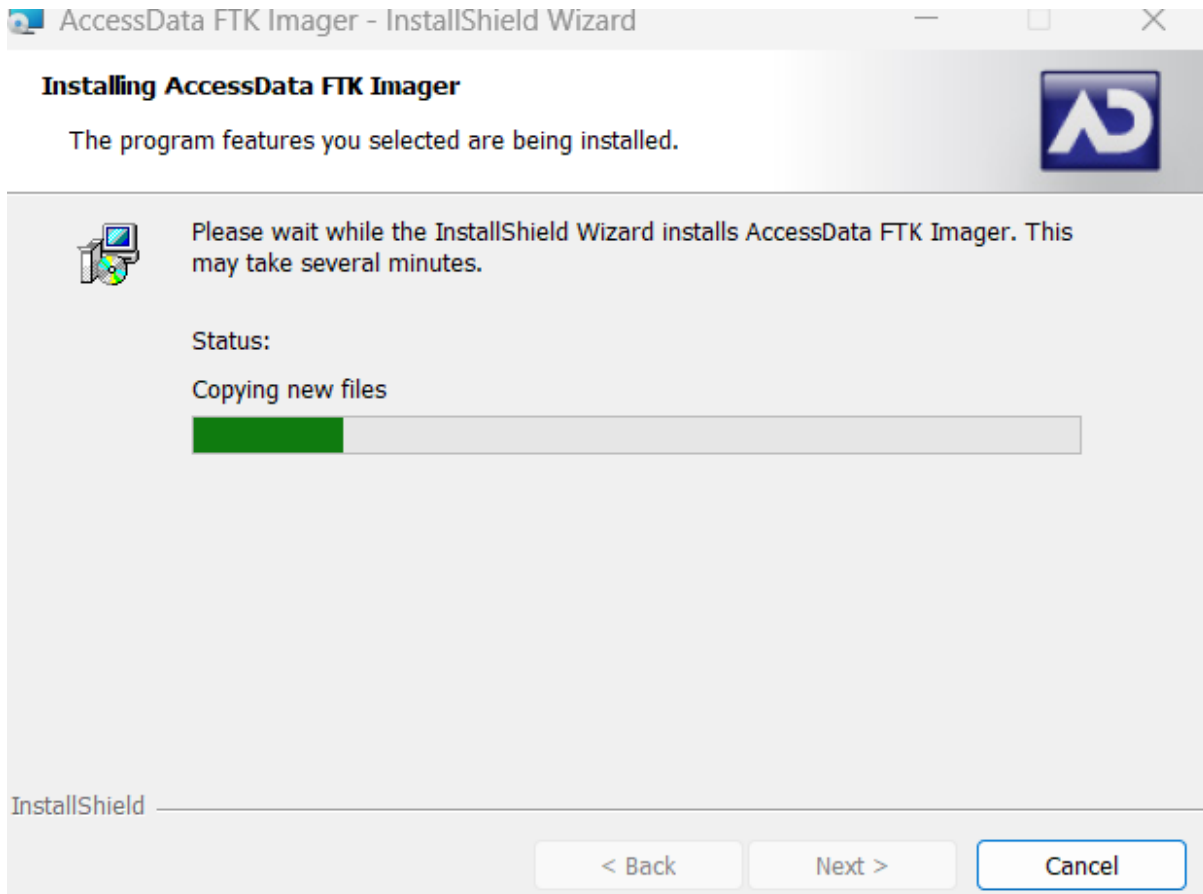
Hash verification confirmed no evidence tampering.

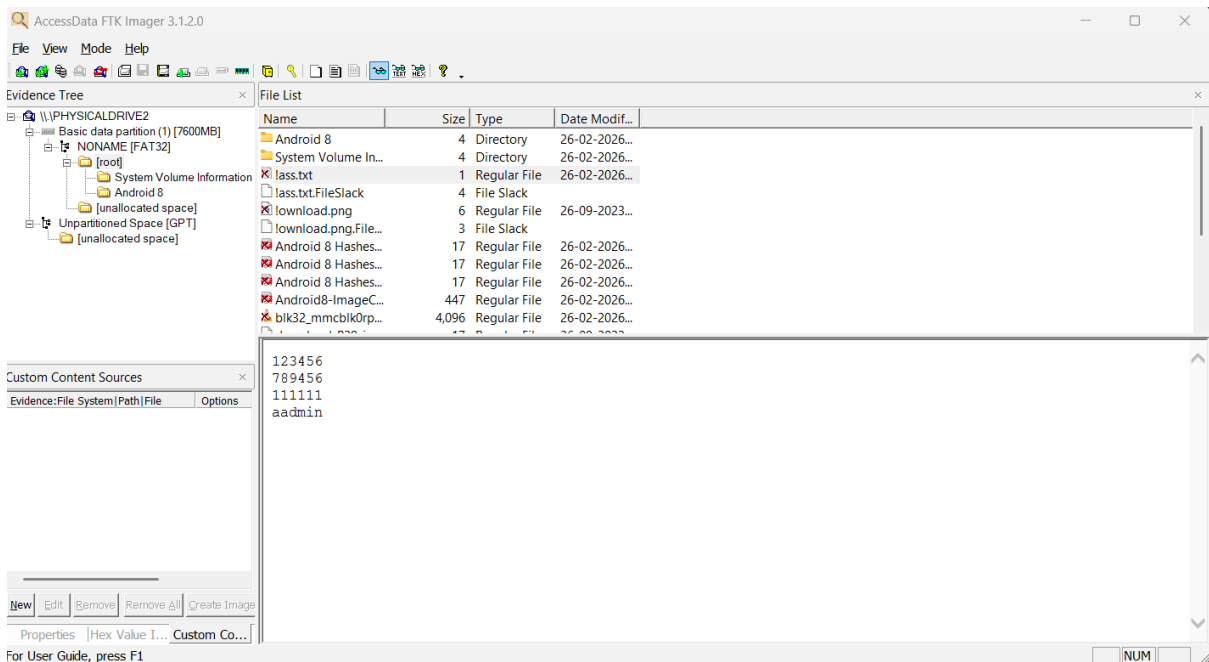
Registry and network artifacts correlated strongly, suggesting data exfiltration attempt.

The methodology followed industry standards ensuring legal admissibility and technical reliability.

30. Screenshots:









AccessData FTK Imager 3.12.0

File View Mode Help

Evidence Tree

- \\PHYSICALDRIVE2
 - Basic data partition (1) [7600MB]
 - NONAME (FAT32)
 - [root]
 - System Volume Information
 - Android 8
 - [unallocated space]
 - Unpartitioned Space [GPT]
 - [unallocated space]

File List

Name	Size	Type	Date Modified
Android 8 Hashes...	17	Regular File	26-02-2026..
Android8-ImageC...	447	Regular File	26-02-2026..
bik32_mmcblk0rp...	4096	Regular File	26-02-2026..
download_P29_jc...	17	Regular File	26-09-2023..
download_P29_jc...	4	File Slack	
Forensic_toolkit_d...	101	Regular File	26-05-2025..
Forensic_toolkit_d...	4	File Slack	
hidden.gif	48	Regular File	18-10-2023..
LG_GSM_H790 Ne...	2	Regular File	04-04-2019..
LG_GSM_H790 Ne...	3	File Slack	
New Microsoft W...	0	Regular File	26-02-2026..

Custom Content Sources

Evidence: File System/Path/File Options

\\PHYSICALDRIVE2/Basic data partition (1) [7600MB]/NONAME (FAT32)/[root]/Forensic_toolkit_dataset.json

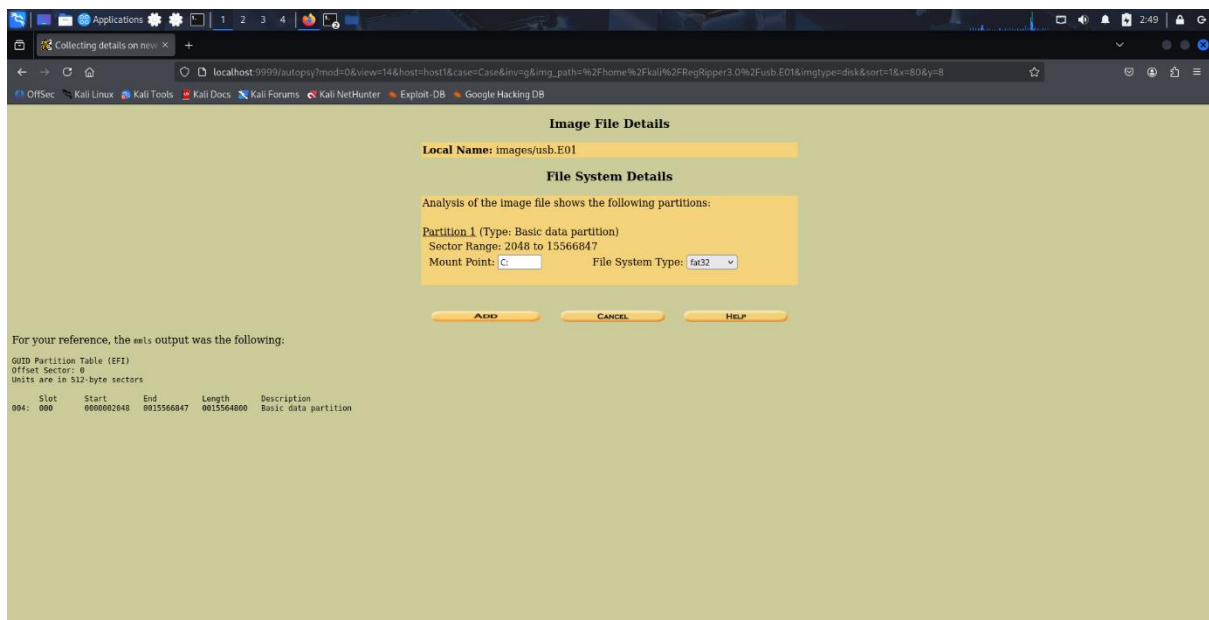
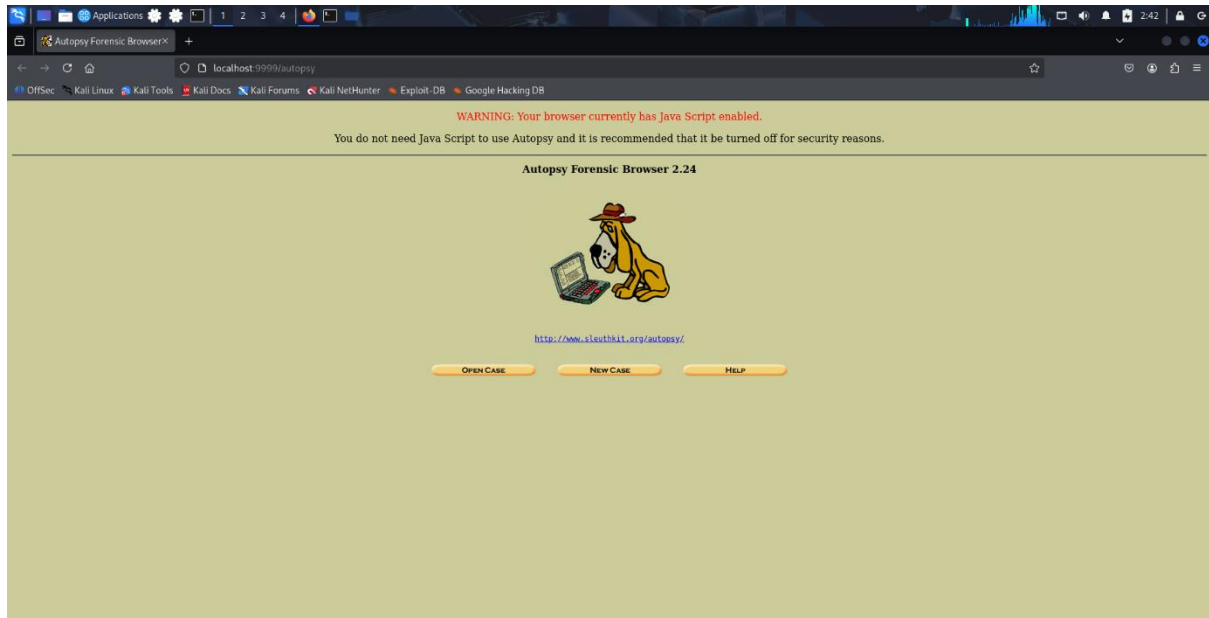
HashMyFiles

File Edit View Options Help

Filename	MD5	SHA1	CRC32	SHA-256	SHA-512	SHA-384	Full
usb image.E01	742328d133d743a6994ea90352f94d9	79b572848bf108af1a1eda0f9a3a2739b42d044	04d730e5	9e9c67874c50a972433cac80cabd2b3c04c0c...	685a5949553c2cd7771144573336fa3b6966b4...	cd4d8f88e55da7a12383ae0a94b67bc3011e2...	C:\

1 file(s)

NirSoft Freeware. <https://www.nirsoft.net>





Case: host1vol4

localhost:9999/autopsy/case=Case&host=host1&img&vol=vol4&mod=1&submod=5&block=32800&len=208

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FILE ANALYSIS KEYWORD SEARCH FILE TYPE IMAGE DETAILS META DATA DATA UNIT HELP CLOSE

PREVIOUS NEXT EXPORT CONTENTS ADD NOTE

Sector Number:
32800

Number of Sectors:
1

Sector Size: 512

Address Type:
Regular (dd)

Lazarus Addr: ☐

VIEW

ALLOCATION LIST

Sectors: 32800-33007
Status: Allocated
[Hide Meta Data Address](#)
Pointed to by Dir Entry: 9
Pointed to by file:

ASCII Contents of Sectors 32800-33007 in usb.E02-2848-15566847

```
{
  "id": 1, "tool_name": "The Sleuth Kit (TSK)", "commands": [
    "ls -l / image.dd > bodyfile",
    "ls -l image.dd",
    "icat image.dd 12345 > output.file",
    "istat image.dd 12345"
  ], "usage": "Analyze disk images to recover files, list file metadata, and create timelines.", "description": "Open-source collection of command-line tools for analyzing disk images and file systems (NTFS, FAT, ext). Enables recovery of deleted files, metadata examination, and timeline generation.", "link": "https://www.sleuthkit.org/sleuthkit/", "system": ["Linux", "Windows", "macOS"]
}, {
  "id": 2, "tool_name": "Autopsy", "commands": [
    "autopsy -i image.dd -o case output"
  ], "usage": "GUI-based, no direct commands for analysis", "usage": "Graphical interface for analyzing disk images, recovering files, and generating reports.", "description": "GUI for The Sleuth Kit, supporting timeline analysis, keyword searching, file recovery, and module integration for detecting IOCs.", "link": "https://www.autopsy.com/", "system": ["Linux", "Windows", "macOS"]
}, {
  "id": 3, "tool_name": "FTK Imager", "commands": [
    "ftkimage image.dd output dir -e01",
    "ftkimage -verify image.e01"
  ], "usage": "Create and verify forensic disk images without altering evidence.", "description": "Free tool for creating forensic images in E01, E02, or A01 formats. Calculates MD5/SHA1 hashes and allows file preview and recovery.", "link": "https://www.exterro.com/ftk-imager/", "system": ["Windows"]
}, {
  "id": 4, "tool_name": "Wireshark", "commands": [
    "wireshark -k -t 0.000000 -r http_request"
  ], "usage": "Capture and analyze network traffic for anomalies or malicious activity.", "description": "Open-source network protocol analyzer for capturing and dissecting packets, supporting over 2,000 protocols.", "link": "https://www.wireshark.org/", "system": ["Linux", "Windows", "macOS"]
}, {
  "id": 5, "tool_name": "Volatility", "commands": [
    "volatility -f memdump.dmp --profile=Win7SP1x86 pslist",
    "volatility -f memdump.dmp --profile=Win7SP1x86 netscan"
  ], "usage": "Analyze memory dumps to extract processes, network connections, and artifacts.", "description": "Memory forensics framework for analyzing RAM dumps, extracting processes, network connections, and keys.", "link": "https://www.volatilityfoundation.org/", "system": ["Linux", "Windows", "macOS"]
}, {
  "id": 6, "tool_name": "Encase", "commands": [
    "encase -i image.dd -o case output"
  ], "usage": "Perform comprehensive forensic analysis of digital evidence.", "description": "Commercial forensic suite for disk imaging, file recovery, and detailed analysis with reporting capabilities.", "link": "https://www.opentext.com/products/encase-forensic/", "system": ["Windows"]
}, {
  "id": 7, "tool_name": "X-Ways Forensics", "commands": [
    "xways -i image.dd"
  ], "usage": "Analyze disk images and memory dumps with advanced file carving.", "description": "Commercial forensic tool for in-depth file system analysis, data recovery, and evidence reporting.", "link": "http://www.x-ways.net/forensic/", "system": ["Windows"]
}, {
  "id": 8, "tool_name": "Caine", "commands": [
    "caine -start"
  ], "usage": "Bootable Linux distro for forensic analysis.", "description": "Open-source forensic environment with pre-installed tools for disk imaging, memory analysis, and reporting.", "link": "https://www.caine-linux.net/", "system": ["Linux"]
}, {
  "id": 9, "tool_name": "Magnet AXIOM", "commands": [
    "magnet -i image.dd"
  ], "usage": "Analyze digital evidence from multiple sources with a unified interface.", "description": "Commercial tool for acquiring and analyzing data from computers, mobiles, and cloud services.", "link": "https://www.magnetforensics.com/products/magnet-axiom/", "system": ["Windows", "macOS"]
}, {
  "id": 10, "tool_name": "Rekal", "commands": [
    "rekall -f memdump.dmp pslist",
    "rekall -f memdump.dmp netscan"
  ], "usage": "Analyze memory dumps for forensic artifacts.", "description": "Open-source memory forensics framework, alternative to Volatility, with plugin-based analysis.", "link": "http://www.rekal-forensic.com/", "system": ["Linux", "Windows", "macOS"]
}, {
  "id": 11, "tool_name": "dd", "commands": [
    "dd if=/dev/zero of=image.dd bs=4k conv=noerror,sync"
  ], "usage": "Create raw disk images for forensic analysis.", "description": "Unix utility for low-level disk copying and imaging, widely used for forensic acquisition.", "link": "https://man7.org/linux/man-pages/dd.1.html", "system": ["Linux", "macOS"]
}, {
  "id": 12, "tool_name": "Foremost", "commands": [
    "foremost -i image.dd -o output dir"
  ], "usage": "Recover files from disk images based on file headers.", "description": "Open-source tool for file carving, recovering files by identifying headers and footers.", "link": "http://foremost.sourceforge.net/", "system": ["Linux", "Windows", "macOS"]
}, {
  "id": 13, "tool_name": "Bulk Extractor", "commands": [
    "bulk_extractor -o output dir image.dd"
  ], "usage": "Extract structured data like emails and URLs from disk images.", "description": "Tool for extracting features (e.g., emails, URLs, credit card numbers) from forensic images.", "link": "https://github.com/simsong/bulk_extractor", "system": ["Linux", "Windows", "macOS"]
}, {
  "id": 14, "tool_name": "Scalpel", "commands": [
    "scalpel -c scalpel.conf -o output dir image.dd"
  ], "usage": "Carve files from disk images using header/footer matching.", "description": "Open-source file carving tool, optimized for speed and efficiency in data recovery.", "link": "https://github.com/sleuthkit/scalpel", "system": ["Linux", "Windows", "macOS"]
}, {
  "id": 15, "tool_name": "TestDisk", "commands": [
    "testdisk image.dd"
  ], "usage": "Recover lost partitions and repair filesystems.", "description": "Open-source tool for recovering lost partitions and making non-booting disks bootable.", "link": "https://www.cgsecurity.org/wiki/TestDisk", "system": ["Linux", "Windows", "macOS"]
}, {
  "id": 16, "tool_name": "PhotoRec", "commands": [
    "photorec image.dd"
  ], "usage": "Recover files from damaged or formatted storage.", "description": "Open-source tool for recovering files based on file signatures,
```

Case: host1vol4

localhost:9999/autopsy/mod=1&submod=3&case=Case&host=host1&img&vol=vol4

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FILE ANALYSIS KEYWORD SEARCH FILE TYPE IMAGE DETAILS META DATA DATA UNIT HELP CLOSE

PREVIOUS NEXT REPORT VIEW CONTENTS EXPORT CONTENTS ADD NOTE

Dir Entry Number:

VIEW

ALLOCATION LIST

Search for File Name

File Type (Recovered):
GIF image data, version 87a, 1340 x 598
MD5 of recovered content:
45ad01e4661848ac2699ac5b38e754

SHA-1 of recovered content:
68862645e3445581e28047273d11f436364c3

Details:
Directory Entry: 27
Not Allocated
File Attributes: File, Archive
Size: 17069
Name: _NDROI-1.PDF
Directory Entry Times:
Written: 2026-02-26 10:35:40 (EST)
Accessed: 2026-02-26 00:00:00 (EST)
Created: 2026-02-26 10:35:39 (EST)
Sectors:
33024 33025 33026 33027 33028 33029 33030 33031



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